

Daniel Williford

Applied AI Automation Systems Engineer

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PROFESSIONAL SUMMARY

Applied AI automation systems engineer with enterprise IT roots and 9 years building independent finance research, data, dashboard, and automation systems. I build and operate automation for evidence heavy workflows, with AI used as a research, coordination, and review layer rather than an uncontrolled decision maker. Strong fit for applied AI, automation specialist, forward deployed engineering, AI infrastructure, AI solutions, and CloudOps/DevOps roles.

CORE SKILLS

Automation and workflow systems: workflow discovery, acceptance criteria, service boundaries, typed packets, runbooks, dashboards, approval steps, exception paths, handoff artifacts.

Agentic AI systems: tool using agents, role scoped context, RAG, structured outputs, LangGraph style workflows, eval checks, trace artifacts, human in the loop review.

Backend and dashboards: Python, FastAPI style APIs, Pydantic contracts, React/TanStack dashboards, CLI automation, JSON/YAML artifacts, Docker Compose demos, pytest validation.

Data and research systems: PostgreSQL, SQLite, PyArrow, Parquet, embeddings, vector search, source ledgers, data cleaning, freshness checks, schema validation, reproducible artifacts.

AI infrastructure operations: Linux services, local LLM endpoints, hosted API fallback, small model worker routing, systemd, journald, health checks, controlled restarts, VPN secured access, GPU workload awareness.

Safety and observability: allowlisted tools, dry run defaults, path boundary checks, audit traces, approval records, logs, eval regressions, telemetry, production mutation guards.

PROFESSIONAL EXPERIENCE

Independent Automation, Finance Research, and AI Systems Developer | 2017 Present

- Built and operated internal automation systems for finance research, data ingestion, data cleaning, backtesting, dashboards, reporting, alerts, and review workflows.
- Developed human in the loop research tools where AI assists with evidence gathering, summarization, candidate ranking, paper trial monitoring, plots, and inspection packets while final decisions stay human owned.
- Maintained a small VPN secured services network supporting several users and multiple machines, including local AI services, tool contracts, file services, dashboards, and controlled service lifecycle management.
- Added validation discipline through source freshness checks, leakage checks, schema validation, eval harnesses, trace artifacts, and explicit no broker execution boundaries.
- Operated cost aware local infrastructure for LLMs, analytics, dashboards, research automation, and GPU workloads.

Independent Video Production and Client Services | 2013 2020

- Ran client facing production work with scheduling, technical setup, media workflows, delivery requirements, customer communication, vendor coordination, and deadline driven execution.
- Transitioned full time into finance research systems and automation after 2020 as finance and software automation became the primary business focus.

IT Specialist, CSC / Raytheon | 2011 2013

- Supported enterprise IT across offices, labs, and restricted access environments.
- Performed installs, upgrades, break/fix troubleshooting, access checks, and network connectivity triage.
- Followed standardized configuration and validation procedures while coordinating with Linux administrators and network teams.

IT Technician, Unified School District 453 | 2009 2011

- Installed and maintained end user systems, supported helpdesk workflows, handled basic network troubleshooting, and improved documentation for repeat issues.

SELECTED SYSTEMS EVIDENCE

Human in the Loop ML Finance Research Copilot

- Built advisory research dashboard and agent workflow support for market/feed ingestion, ML validation, backtest review, source freshness, paper trial monitoring, candidate ranking, plots, and human promotion decisions.

Agentic AI Ops

- Built local FastAPI + LangGraph workflow service with typed packets, traces, artifacts, approval controls, dry run behavior, source backed packets, and tests.

Agentic Harnesses and Self Hosted AI Services

- Built role profiles, profile memory/config, allowed tools, small local model workers, larger model synthesis, hosted API fallback, A and B variants, tool call telemetry, cost aware routing, local LLM endpoints, and VPN secured Linux/GPU services.

Evidence Backed Research Pipeline

- Built source grading, evidence ledgers, critique loops, persistent research artifacts, and sample research packets for auditable research workflows.

CERTIFICATIONS AND TRAINING

CompTIA A+ | CompTIA Network+

Selected coursework: HarvardX/edX CS50AI, CS50SQL, MITx Foundations of Modern Finance, Machine Learning with Python.